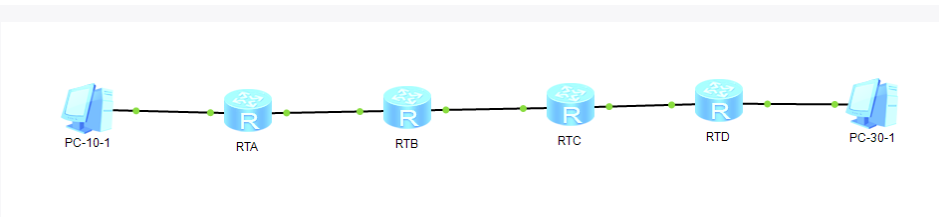


Lab-5 (1) Basic RIP Configuration and Analysis(E)

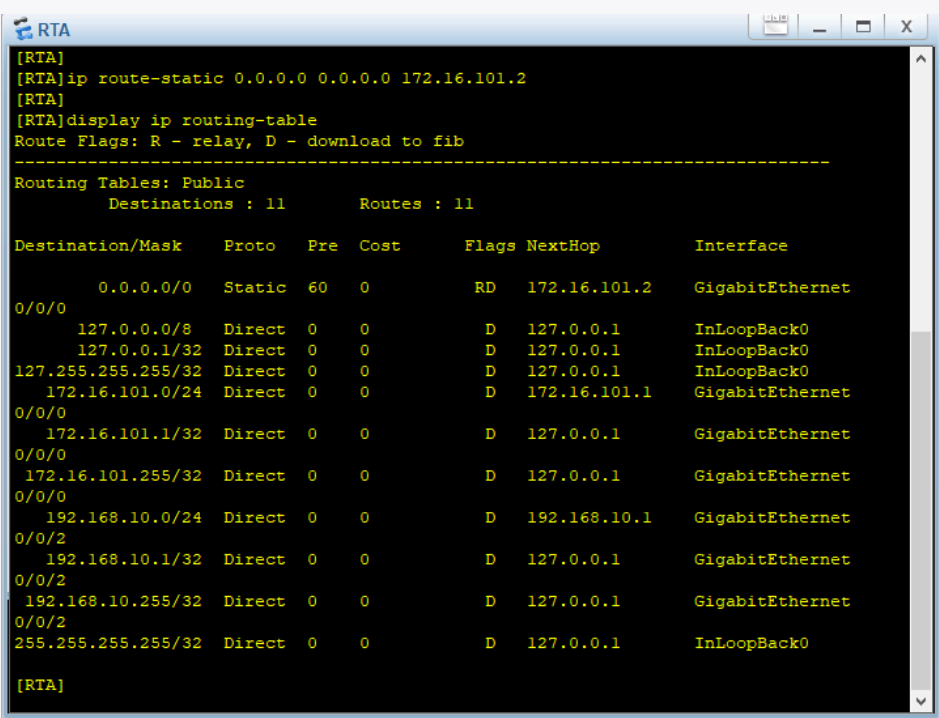
07112303 左逸龙 1120231863
2026-05-22

Lab-5-1 BASIC RIPV1 CONFIGURATION

Q5-1.1. Paste the screenshot of the created topology.

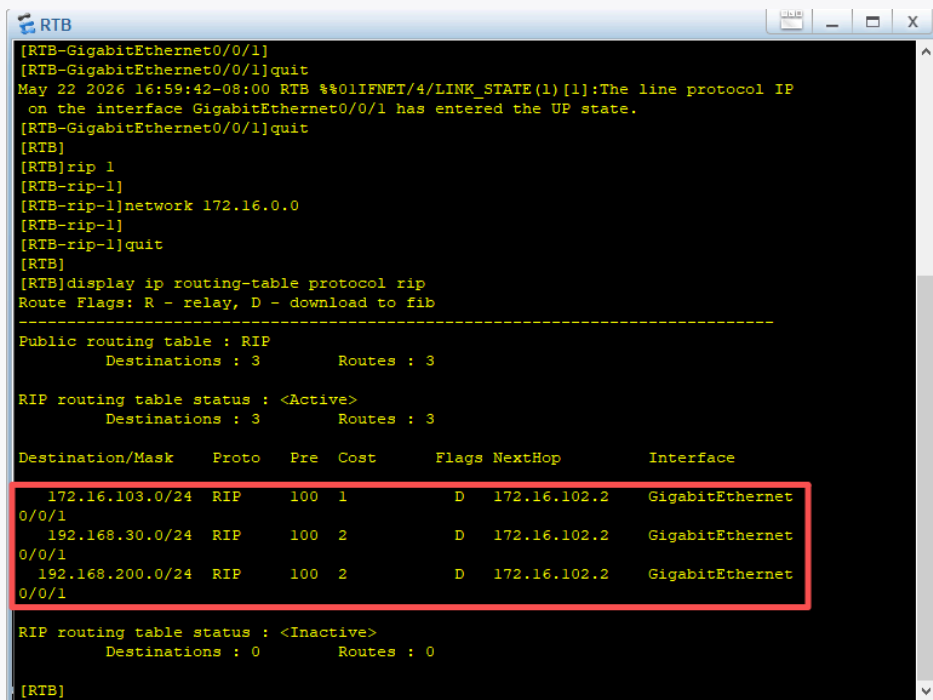


Q5-1.2. Paste the screenshot of the IP routing table on the router RTA. Highlight the RIPv1 routes in the screenshot. Which routes are learned from the RIP neighbors?



RTA 并没有从 RIP 邻居学习到任何路由。由于 RTA 仅配置了去往 0.0.0.0/0 的静态默认路由 (Static)，且没有运行 RIP 协议，因此其路由表中不存在通过 RIP 动态学习到的路由。

Q5-1.3. Paste the screenshot of the IP routing table on the router RTB. Highlight the RIPv1 routes in the screenshot. Which routes are learned from the RIP neighbors?



```
[RTB-GigabitEthernet0/0/1]
[RTB-GigabitEthernet0/0/1]quit
May 22 2026 16:59:42-08:00 RTB %%01IFNET/4/LINK_STATE(1)[1]:The line protocol IP
on the interface GigabitEthernet0/0/1 has entered the UP state.
[RTB-GigabitEthernet0/0/1]quit
[RTB]
[RTB]rip 1
[RTB-rip-1]
[RTB-rip-1]network 172.16.0.0
[RTB-rip-1]
[RTB-rip-1]quit
[RTB]
[RTB]display ip routing-table protocol rip
Route Flags: R - relay, D - download to fib
-----
Public routing table : RIP
Destinations : 3          Routes : 3

RIP routing table status : <Active>
Destinations : 3          Routes : 3

Destination/Mask    Proto    Pre  Cost    Flags NextHop          Interface
-----
172.16.103.0/24    RIP      100   1       D   172.16.102.2      GigabitEthernet
0/0/1
192.168.30.0/24    RIP      100   2       D   172.16.102.2      GigabitEthernet
0/0/1
192.168.200.0/24   RIP      100   2       D   172.16.102.2      GigabitEthernet
0/0/1

RIP routing table status : <Inactive>
Destinations : 0          Routes : 0

[RTB]
```

从截图中可以看出，协议类型（Proto）为 RIP 的路由是通过 RIP 邻居学习到的。RTB 学习到的 RIP 路由包括：

- 目的网段 172.16.103.0/24
- 目的网段 192.168.30.0/24
- 目的网段 192.168.200.0/24

Q5-1.4. Paste the screenshot of the IP routing table on the router RTC. Highlight the RIPv1 routes in the screenshot. Which routes are learned from the RIP neighbors?

```

May 22 2026 16:59:51-08:00 RTC %*O1IFNET/4/LINK_STATE(1)[1]:The line protocol IP
on the interface GigabitEthernet0/0/0 has entered the UP state.
[RTC-GigabitEthernet0/0/0]quit
[RTC]
[RTC]rip 1
[RTC-rip-1]
[RTC-rip-1]network 172.16.0.0
[RTC-rip-1]
[RTC-rip-1]quit
[RTC]
[RTC]display ip routing-table protocol rip
Route Flags: R - relay, D - download to fib
-----
Public routing table : RIP
Destinations : 3      Routes : 3

RIP routing table status : <Active>
Destinations : 3      Routes : 3

Destination/Mask    Proto    Pre  Cost    Flags NextHop         Interface
-----
172.16.101.0/24    RIP      100  1        D   172.16.102.1         GigabitEthernet
0/0/1
192.168.30.0/24    RIP      100  1        D   172.16.103.1         GigabitEthernet
0/0/0
192.168.200.0/24   RIP      100  1        D   172.16.103.1         GigabitEthernet
0/0/0

RIP routing table status : <Inactive>
Destinations : 0      Routes : 0

[RTC]

```

从截图中可以看出，RTC 通过 RIP 邻居学习到了以下目的网段的路由：

- 目的网段 172.16.101.0/24
- 目的网段 192.168.30.0/24
- 目的网段 192.168.200.0/24

Q5-1.5. Paste the screenshot of the IP routing table on the router RTD. Highlight the RIPv1 routes in the screenshot. Which routes are learned from the RIP neighbors?

```

[RTD]rip 1
[RTD-rip-1]
[RTD-rip-1]network 172.16.0.0
[RTD-rip-1]
[RTD-rip-1]network 192.168.30.0
[RTD-rip-1]
[RTD-rip-1]network 192.168.200.0
[RTD-rip-1]
[RTD-rip-1]quit
[RTD]
[RTD]display ip routing-table protocol rip
Route Flags: R - relay, D - download to fib
-----
Public routing table : RIP
Destinations : 2      Routes : 2

RIP routing table status : <Active>
Destinations : 2      Routes : 2

Destination/Mask    Proto    Pre  Cost    Flags NextHop         Interface
-----
172.16.101.0/24    RIP      100  2        D   172.16.103.2         GigabitEthernet
0/0/0
172.16.102.0/24    RIP      100  1        D   172.16.103.2         GigabitEthernet
0/0/0

RIP routing table status : <Inactive>
Destinations : 0      Routes : 0

[RTD]

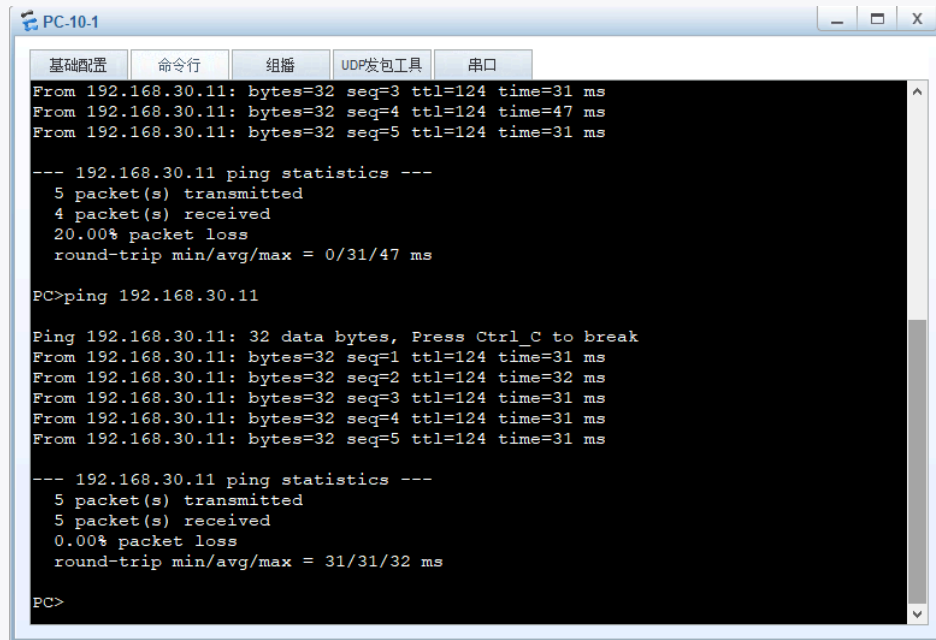
```

从截图中可以看出，RTD 通过 RIP 邻居学习到了以下目的网段的路由：

- 目的网段 172.16.101.0/24
- 目的网段 172.16.102.0/24

Q5-1.6. Can PC-10-1 and PC-30-1 ping each other? Please paste the screenshot of the ping result.

可以互相 ping 通。



```
PC-10-1
基础配置 命令行 组播 UDP发包工具 串口
From 192.168.30.11: bytes=32 seq=3 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=4 ttl=124 time=47 ms
From 192.168.30.11: bytes=32 seq=5 ttl=124 time=31 ms

--- 192.168.30.11 ping statistics ---
 5 packet(s) transmitted
 4 packet(s) received
20.00% packet loss
round-trip min/avg/max = 0/31/47 ms

PC>ping 192.168.30.11

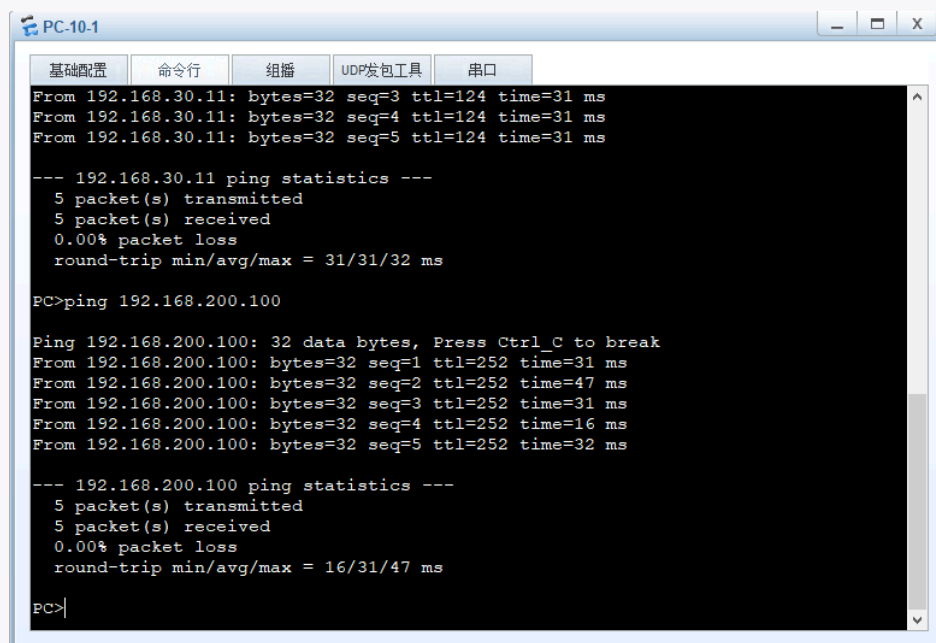
Ping 192.168.30.11: 32 data bytes, Press Ctrl_C to break
From 192.168.30.11: bytes=32 seq=1 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=2 ttl=124 time=32 ms
From 192.168.30.11: bytes=32 seq=3 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=4 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=5 ttl=124 time=31 ms

--- 192.168.30.11 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 31/31/32 ms

PC>
```

Q5-1.7. Can PC-10-1 and loopback 0 on RTD ping each other? Please paste the screenshot of the ping result.

可以互相 ping 通。



```
PC-10-1
基础配置 命令行 组播 UDP发包工具 串口
From 192.168.30.11: bytes=32 seq=3 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=4 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=5 ttl=124 time=31 ms

--- 192.168.30.11 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 31/31/32 ms

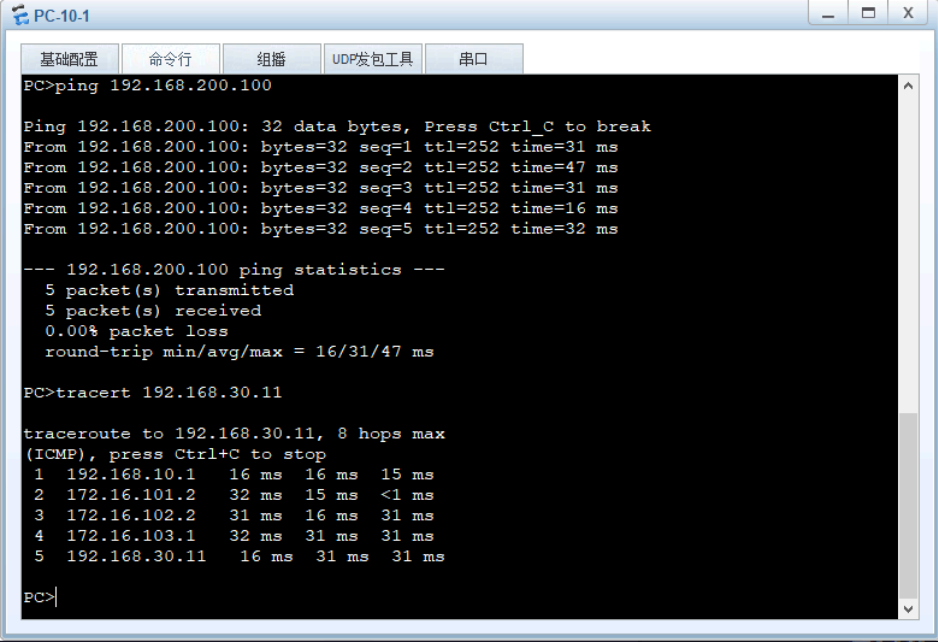
PC>ping 192.168.200.100

Ping 192.168.200.100: 32 data bytes, Press Ctrl_C to break
From 192.168.200.100: bytes=32 seq=1 ttl=252 time=31 ms
From 192.168.200.100: bytes=32 seq=2 ttl=252 time=47 ms
From 192.168.200.100: bytes=32 seq=3 ttl=252 time=31 ms
From 192.168.200.100: bytes=32 seq=4 ttl=252 time=16 ms
From 192.168.200.100: bytes=32 seq=5 ttl=252 time=32 ms

--- 192.168.200.100 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 16/31/47 ms

PC>|
```

Q5-1.8. Please paste the screenshot of the result of command “tracert 192.168.30.11” issued from PC-10-1.



```
PC>ping 192.168.200.100

Ping 192.168.200.100: 32 data bytes, Press Ctrl_C to break
From 192.168.200.100: bytes=32 seq=1 ttl=252 time=31 ms
From 192.168.200.100: bytes=32 seq=2 ttl=252 time=47 ms
From 192.168.200.100: bytes=32 seq=3 ttl=252 time=31 ms
From 192.168.200.100: bytes=32 seq=4 ttl=252 time=16 ms
From 192.168.200.100: bytes=32 seq=5 ttl=252 time=32 ms

--- 192.168.200.100 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 16/31/47 ms

PC>tracert 192.168.30.11

tracert to 192.168.30.11, 8 hops max
(ICMP), press Ctrl+C to stop
 1 192.168.10.1    16 ms  16 ms  15 ms
 2 172.16.101.2   32 ms  15 ms  <1 ms
 3 172.16.102.2   31 ms  16 ms  31 ms
 4 172.16.103.1   32 ms  31 ms  31 ms
 5 192.168.30.11  16 ms  31 ms  31 ms

PC>|
```

Q5-1.9. How many types of RIPv1 packets are there? What are they?

RIPv1 共有两种类型的报文，分别是：

- **Request**（请求报文）：路由器刚启动时或需要更新时主动向邻居发送，用于请求完整的或部分的路由表信息。
- **Response**（响应报文）：路由器周期性广播或在收到 Request 报文后触发发送，包含具体的路由条目信息。

Q5-1.10. What is the interval between RIPv1 routing updates?

RIPv1 默认的路由更新时间间隔是 **30 秒**。

Q5-1.11. Which protocol does RIPv1 use to transmit RIP packets? What are the source port number and destination port number?

RIPv1 使用 **UDP** 协议传输报文。其源端口号和目的端口号均为 **520**。

Q5-1.12. When RIPv1 sends a routing update message, what is the destination IP address of the message? What type of IP address is it?

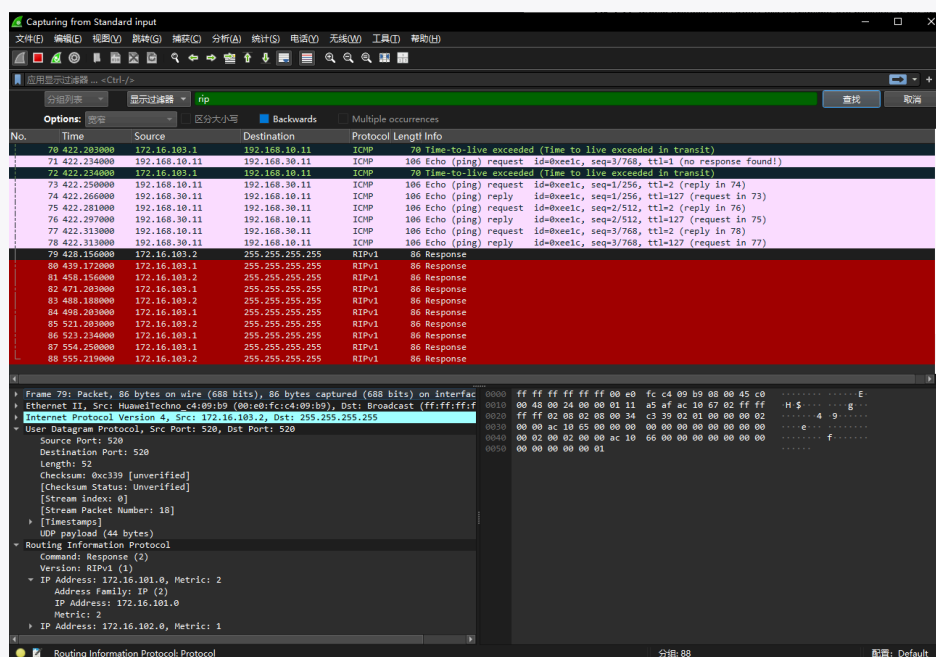
RIPv1 发送路由更新报文时，目的 IP 地址为 255.255.255.255。这是一个受限的**广播地址**。

Q5-1.13. How many routes are there in each RIPv1 route update message? What information does each route contain in the route update message? Please paste the screenshot of the route information contained in the captured update routing message.

每个 RIPv1 路由更新报文最多可以包含 25 条路由记录。

每条路由记录中包含的主要信息有：

- **AFI** (Address Family Identifier, 地址族标识符, IPv4 通常对应的值为 2)
- **IP Address** (目的网络的 IP 地址)
- **Metric** (到达目的网络的跳数开销)



Q5-1.14. Please give the format of RIPv1 routing update message.

RIPv1 更新报文的基本格式如下：

- **Command** (1 字节)：标识报文类型，1 表示 Request，2 表示 Response。
- **Version** (1 字节)：RIP 协议的版本号，对于 RIPv1 此处必须为 1。
- **Must be zero** (2 字节)：保留填充字段，必须全为 0。

随后是最多包含 25 条路由记录 (Route Entries) 的更新列表，每条路由记录长度为 20 字节，具体包含：

- **Address Family Identifier (AFI)** (2 字节)：地址族标识符，IPv4 对应的值为 2。
- **Route Tag / Must be zero** (2 字节)：在 RIPv1 中不使用该标签，必须全为 0。
- **IP Address** (4 字节)：目的网络的 IP 地址。
- **Subnet Mask / Must be zero** (4 字节)：RIPv1 属于有类路由协议，不支持可变长子网掩码 (VLSM)，此字段必须全为 0。

- **Next Hop / Must be zero** (4 字节): RIPv1 中不携带下一跳地址, 此字段必须全为 0。
- **Metric** (4 字节): 到达目的网络的跳数 (Hop Count), 有效取值范围为 1 到 15, 16 表示目标网络不可达。